DAIDP

Building **Agentic AI Applications**

From Models to Agents to Products

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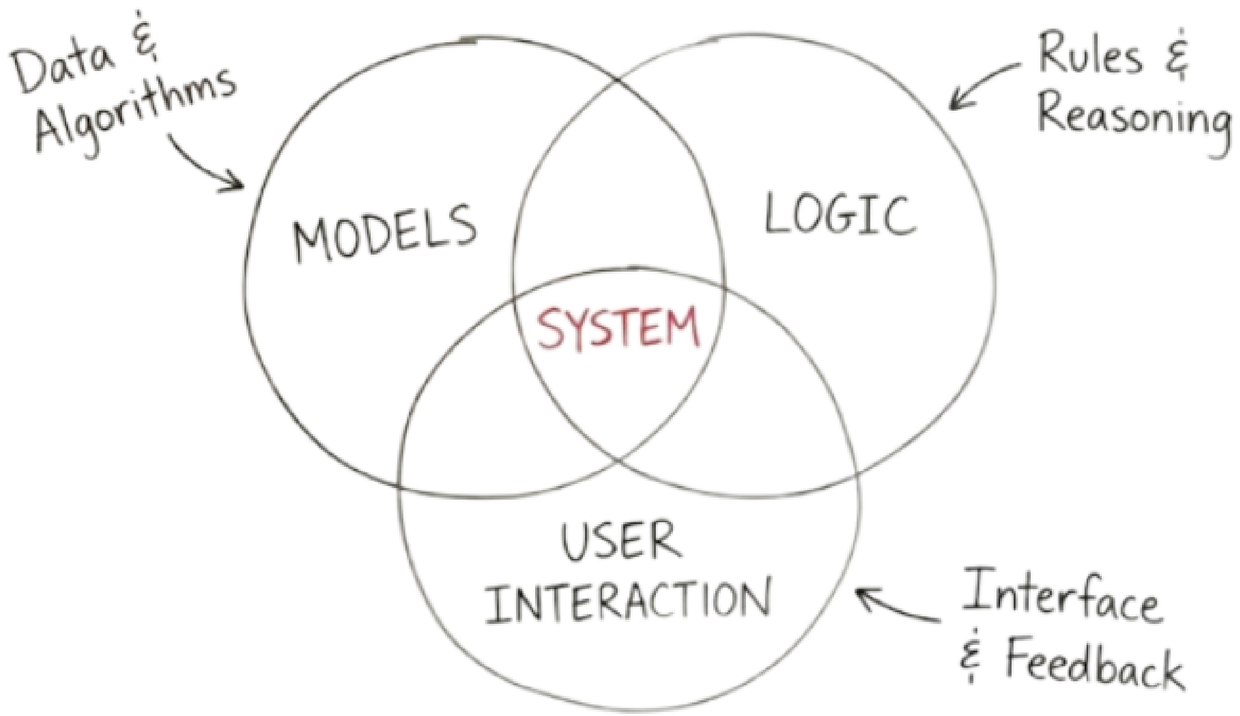
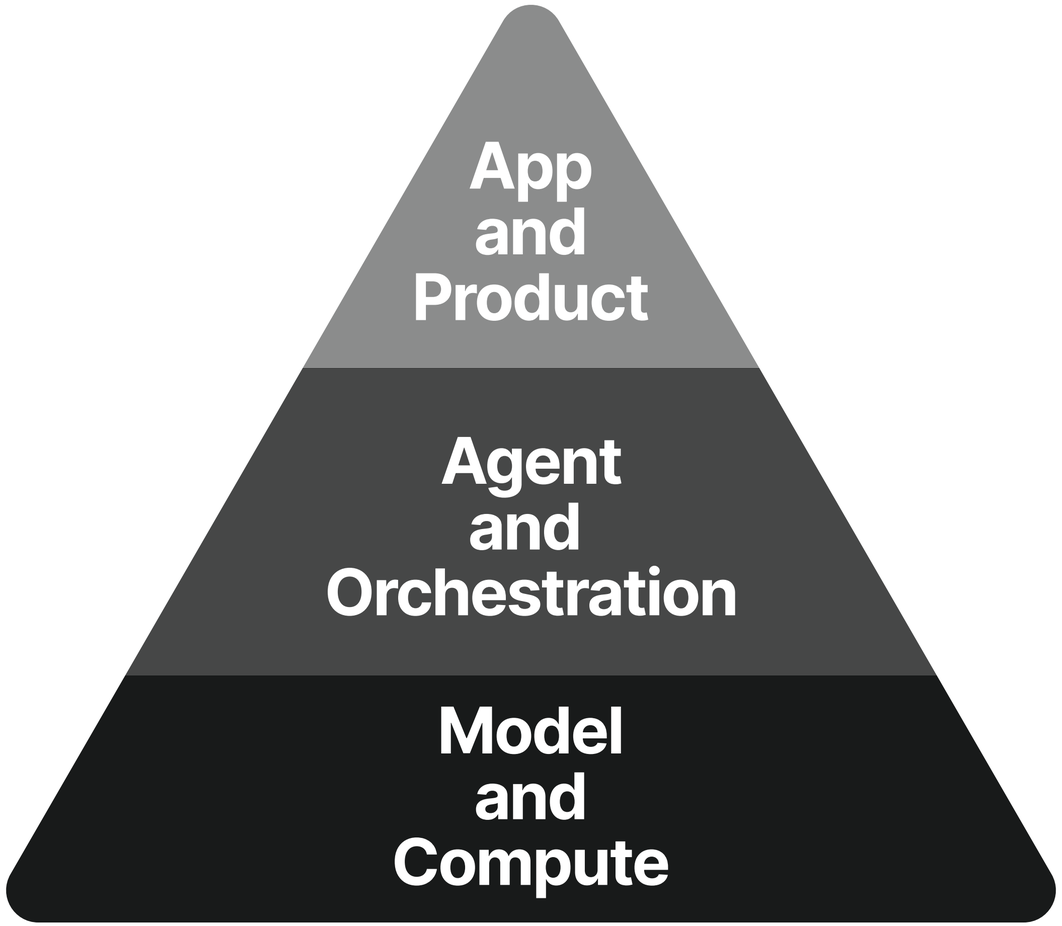
What is an AI-Based Application?

An AI-based application is not just a chatbot.

It is a system that takes input, reasons over it, and decides the best course of action to produce an output that helps achieve a goal.





To understand these systems, we break them into three layers.

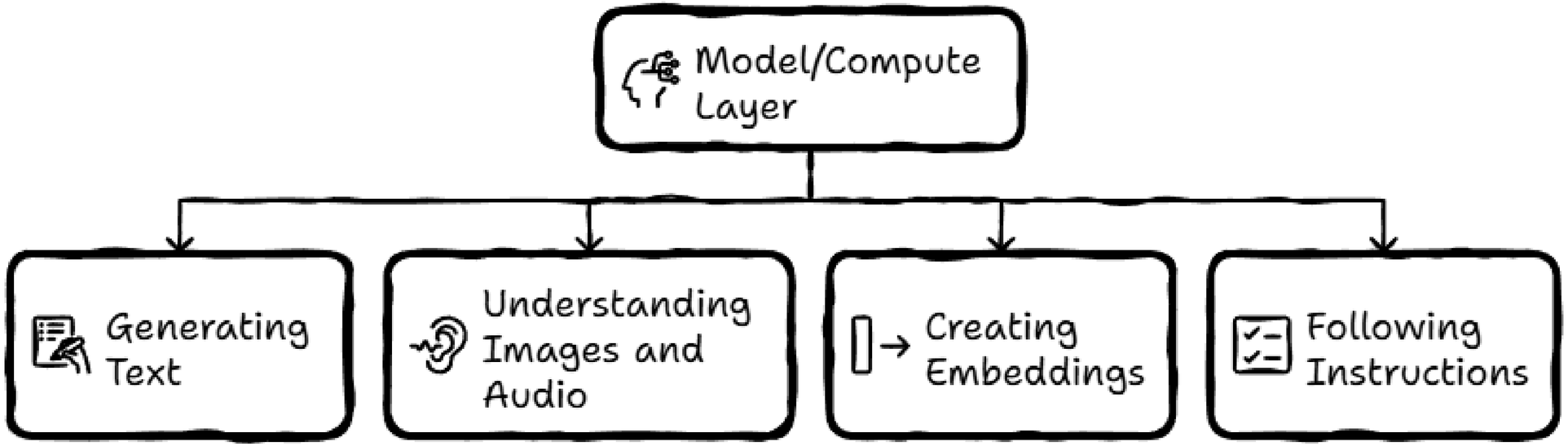
Model / Compute Layer

Agent / Orchestration Layer

App / Product Layer

Model / Compute Layer

The Model / Compute Layer is where AI models live, run and are served either locally or through APIs.



“Where does intelligence come from?”

Some Key Tools

Ollama

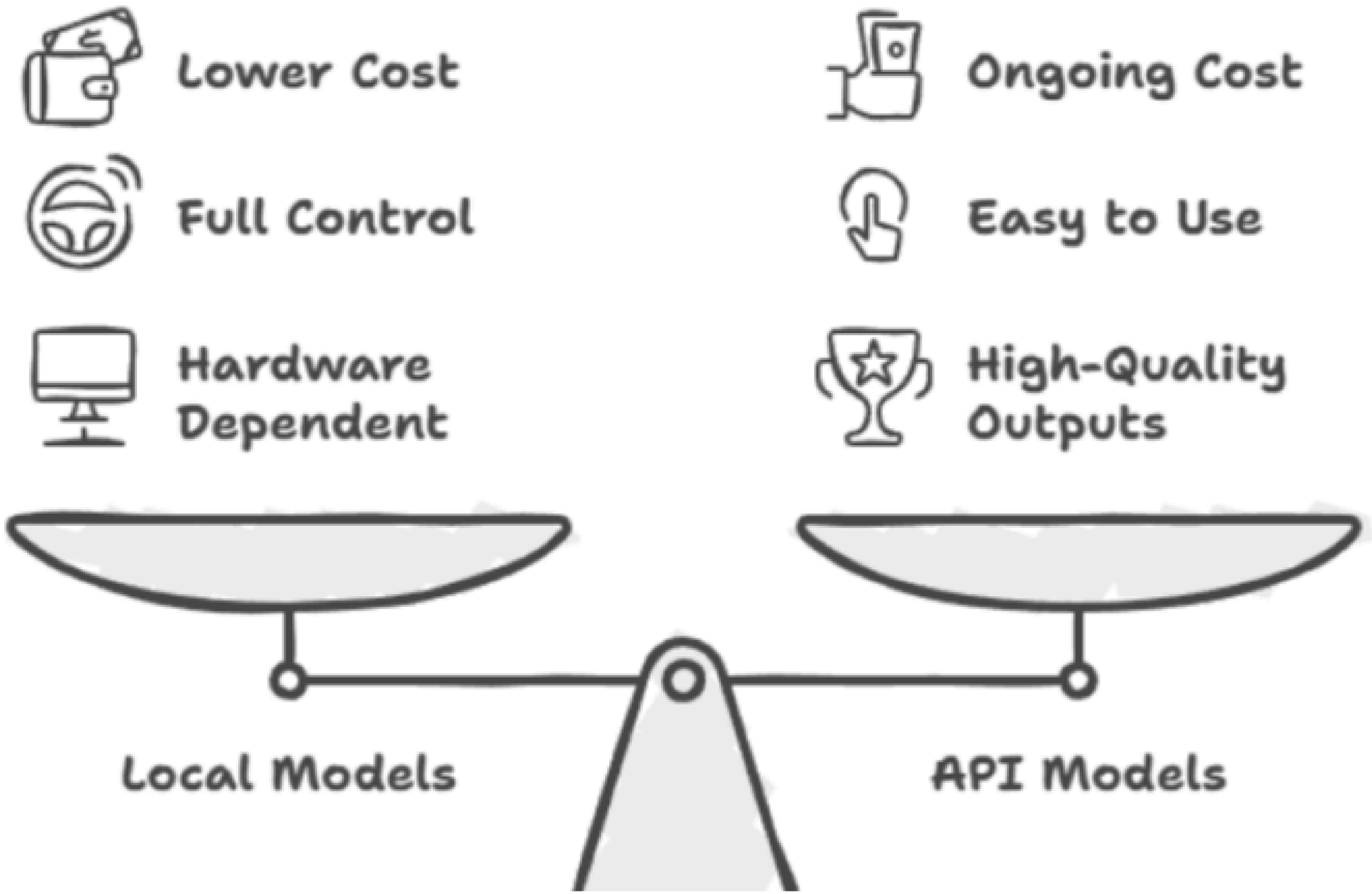
Runs large language models locally with simple commands.

Hugging Face

A library to load, run and fine-tune thousands of open-source models for text, vision and audio tasks.

unsloth

Optimized framework for fine-tuning open-source LLMs efficiently on consumer GPUs.



Choosing between them is a product decision, not just a technical one.

Demo

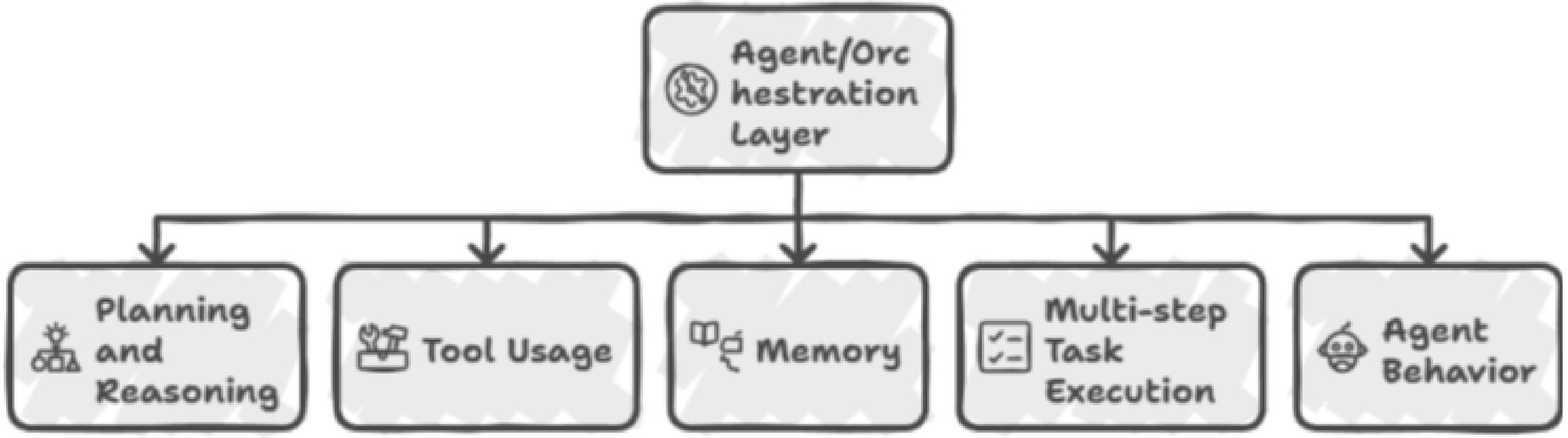
Model / Compute Layer

Agent / Orchestration Layer

App / Product Layer

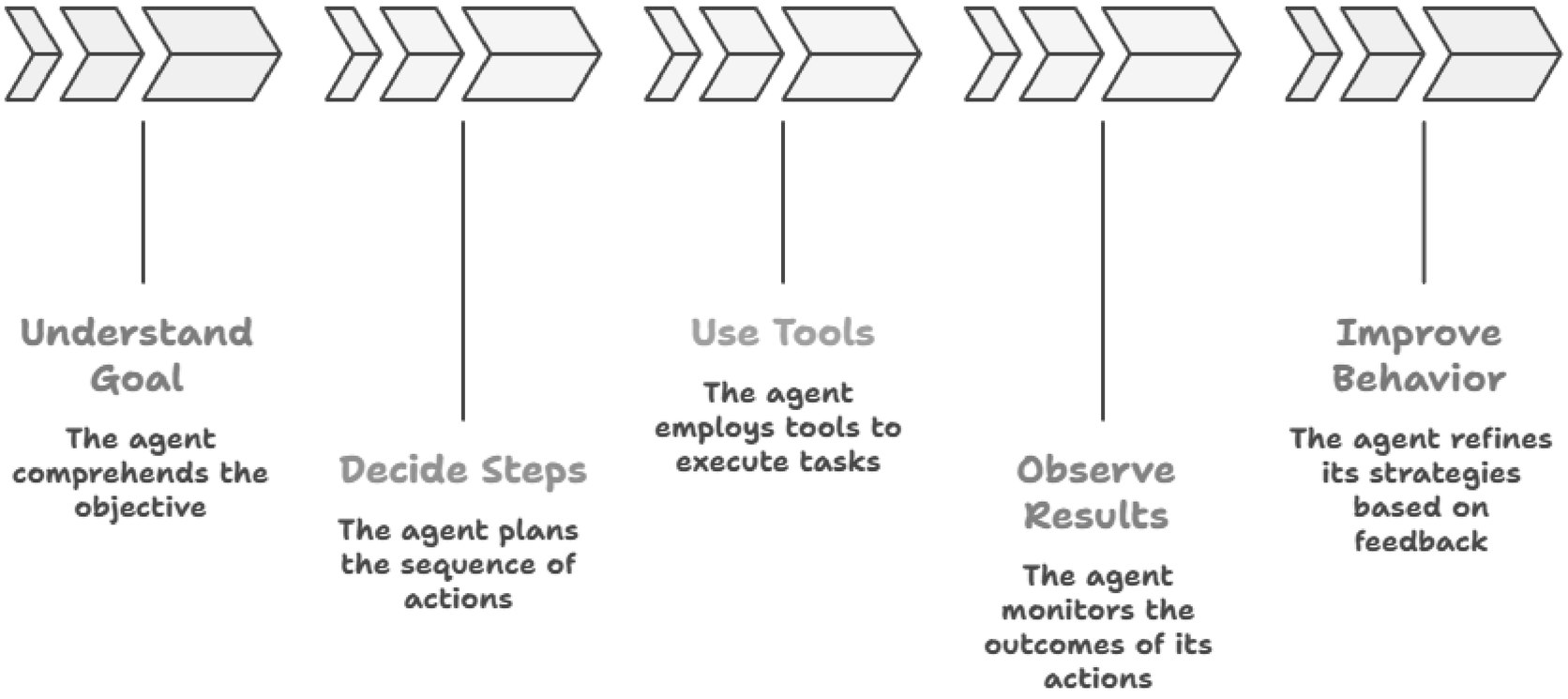
Agent / Orchestration Layer

The Agent / Orchestration Layer layer decides what the AI should do next, why it should do it and how the overall task should be executed.



“What should the AI do next?”

What is an Agent?



A chatbot responds. An agent is an AI system that acts.

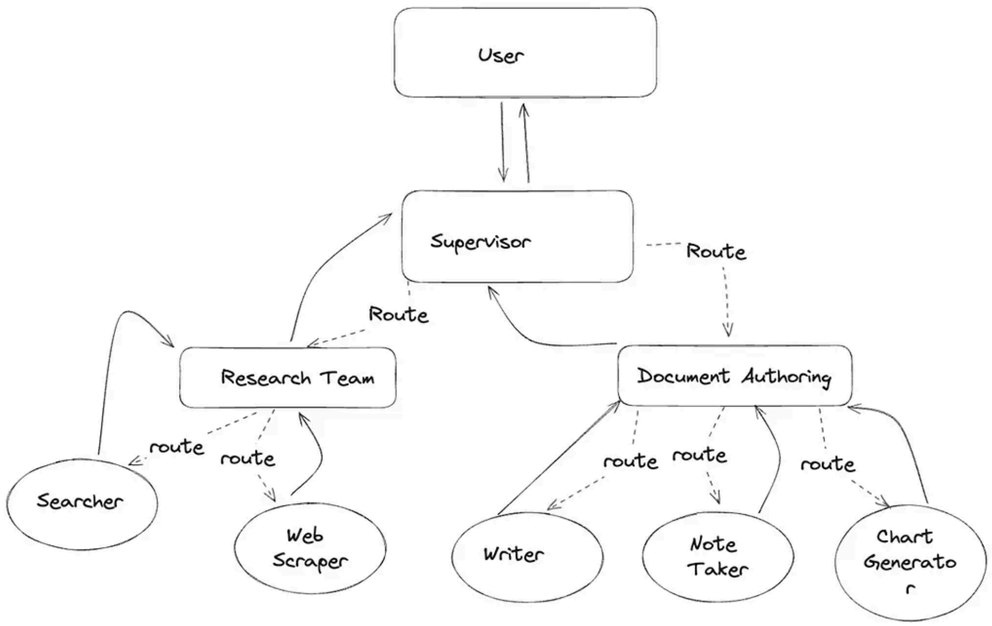
From Single-Agent to Multi-Agent Systems

Agent Orchestration Patterns

When systems grow complex, we need patterns to organize agent behavior.

These patterns define:

- How agents interact
- How tasks flow
- How decisions are made



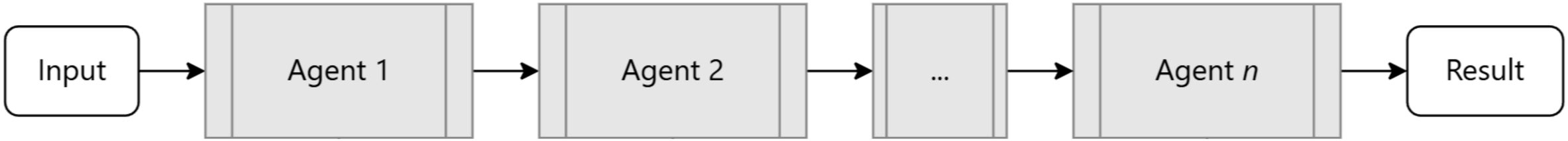
They are architectural patterns, not algorithms.

Sequential Chain Pattern

The output of one agent becomes the input of the next agent.

Used for:

- Data processing pipelines
- Step-by-step analysis
- Research followed by writing

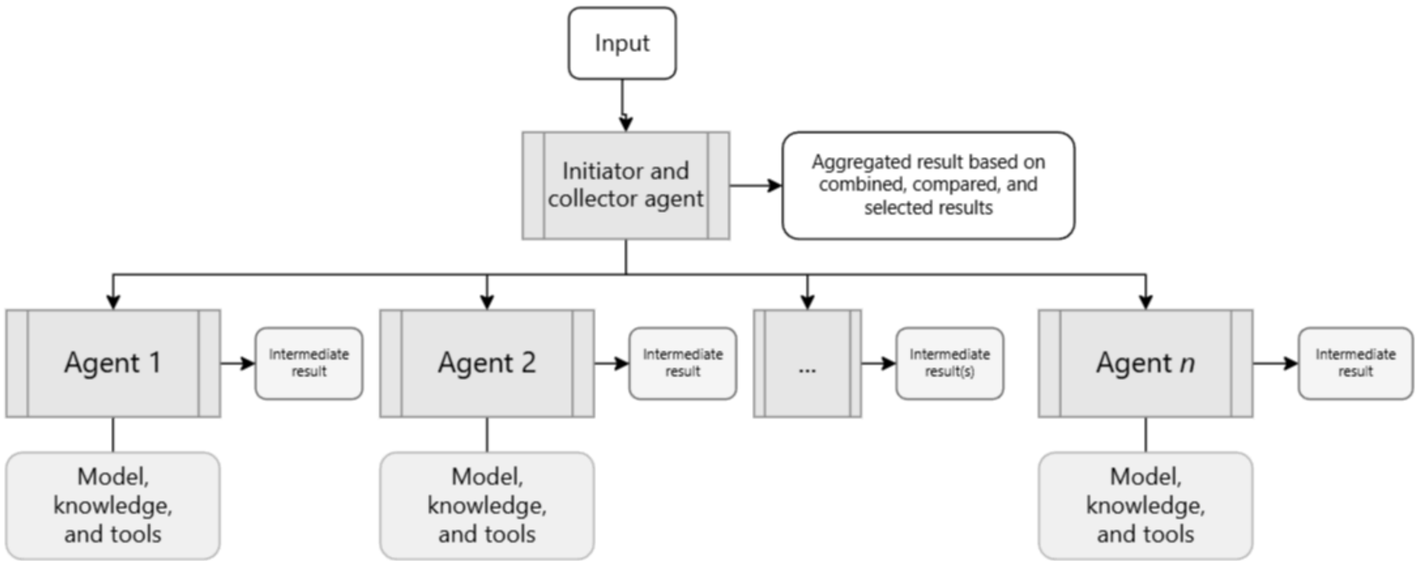


Parallel Execution Pattern

Multiple agents work simultaneously, and their results are combined.

Used for:

- Fact-checking from multiple sources
- Comparing multiple answers
- A/B testing ideas

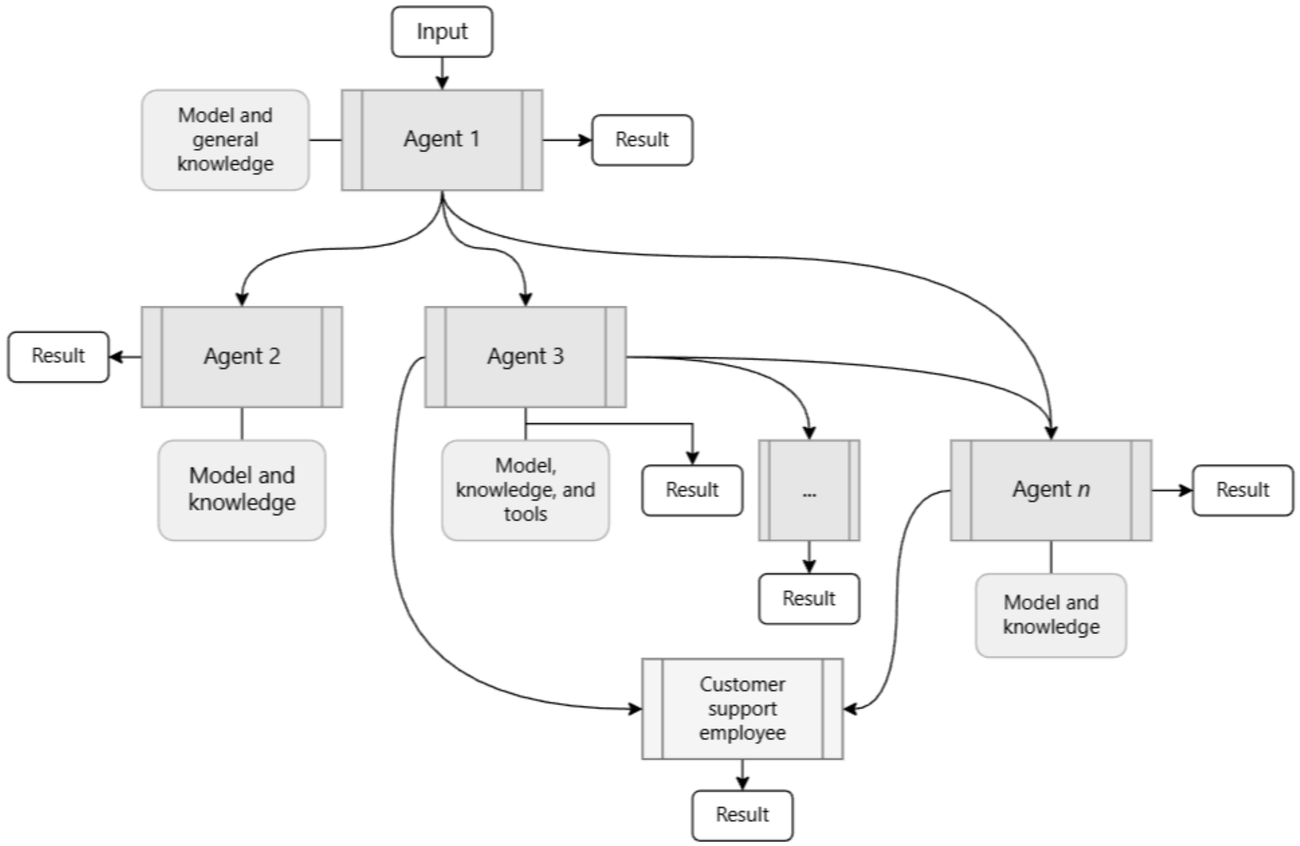


Router Pattern

A central agent routes tasks to specialized agents based on intent.

Used for:

- Customer support systems
- Multi-domain assistants
- Skill-based routing



Enhancing AI Agents

Some Key Tools

LangChain

A framework for building LLM applications using chains, tools, memory and agents.

LangGraph

A graph-based framework for building agent workflows with state, loops and conditions.

Demo

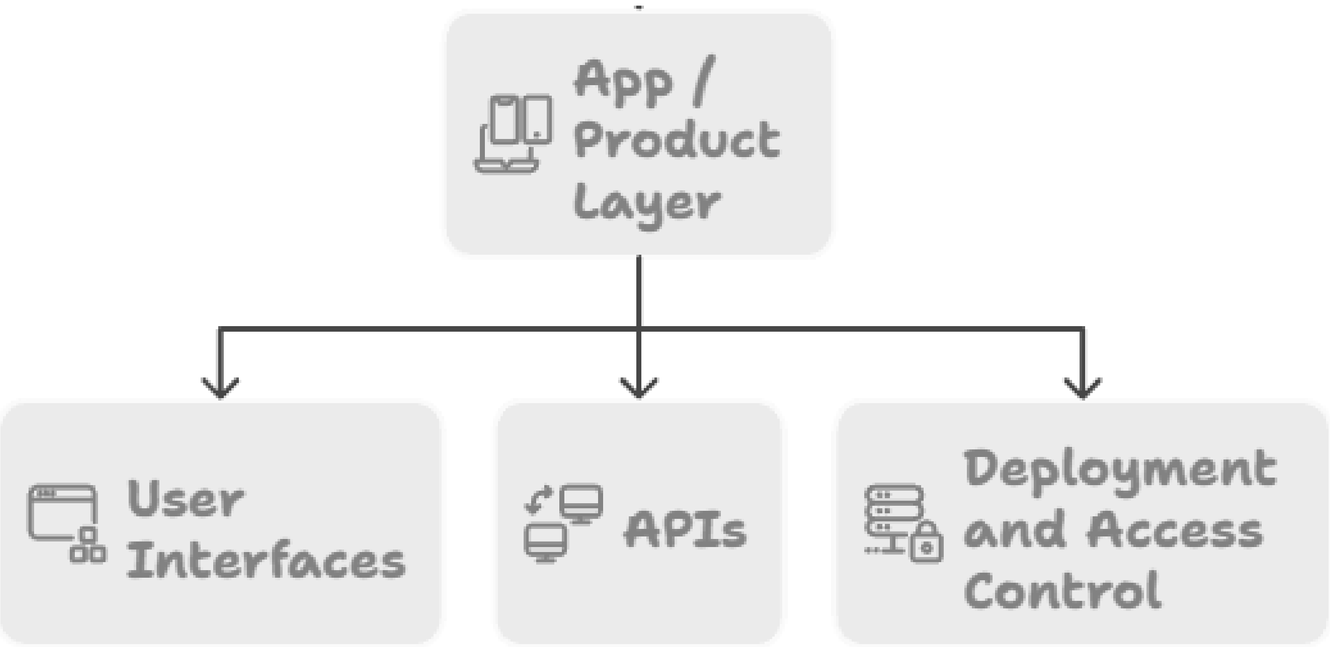
Model / Compute Layer

Agent / Orchestration Layer

App / Product Layer

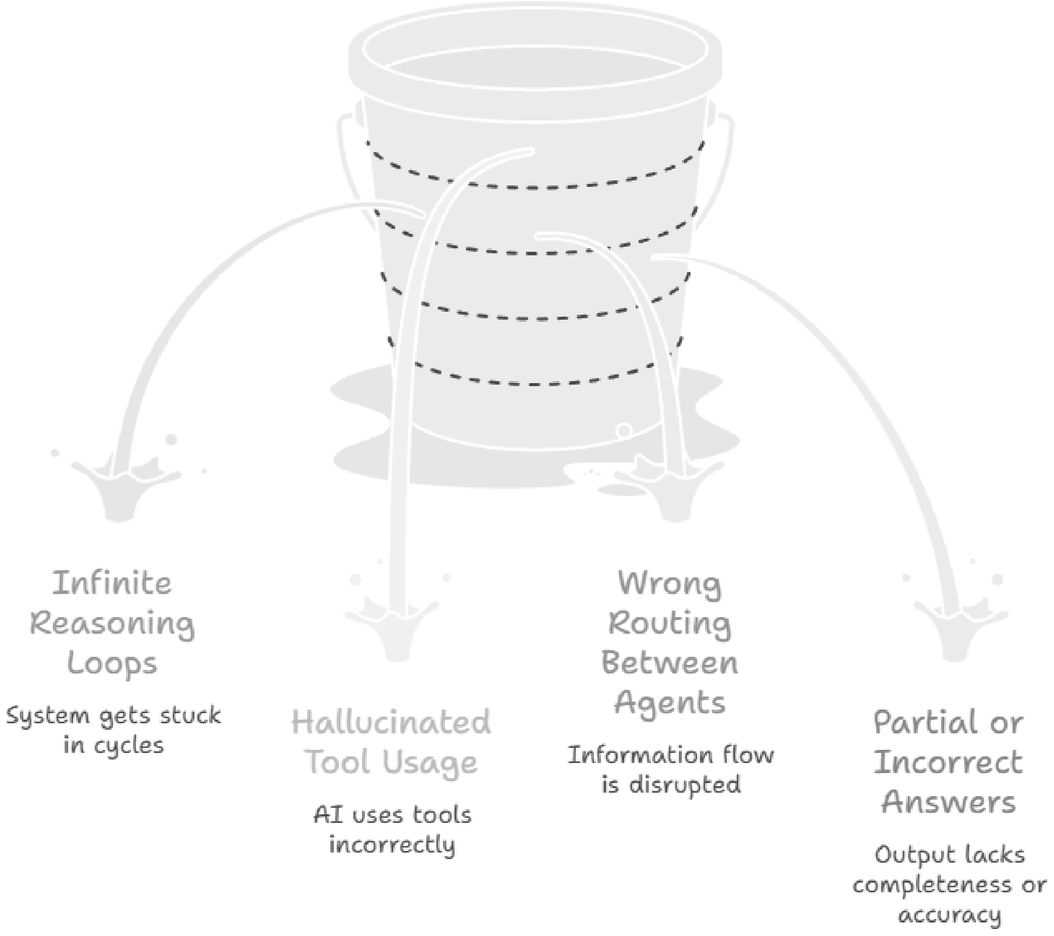
App / Product Layer

The App / Product Layer is where users interact with the AI system.



“How do users experience the AI?”

Why AI Systems Fail?



Making AI Systems Reliable

Limit reasoning steps
Prevent agents from looping endlessly by capping the number of decisions or actions they can take.

Validate outputs
Check whether the agent's output follows the expected format, rules or constraints before accepting it.

Use critic or reviewer agents
Add a secondary agent to review, correct or improve the main agent's output.

Retry on failure
Allow the system to re-attempt a task when the output is incomplete, incorrect or invalid.